

Non-Functional Requirements Document

Digital STEM Content Delivery Platform

Project Details

Field	Details
Project Name	Digital STEM Content Delivery Platform
Client	ABC School
Document Type	Non-Functional Requirements Document
Prepared By	Business Analyst
Version	1.0
Date	28 June 2026

1. Document Purpose

This Non-Functional Requirements Document defines the quality, performance, usability, security, compatibility and operational expectations for the Digital STEM Content Delivery Platform. It supports the Functional Requirements Document by describing how well the system should perform, not only what the system should do.

2. Document Inputs

- Approved Business Requirements Document
- Functional Requirements Document
- As-Is Process Flow
- To-Be Process Flow
- Discovery Meeting Notes
- Teacher, Admin Staff and Principal requirements

3. Scope Summary

The platform will support teacher-led classroom delivery of approved STEM and Coding content through a browser-based web portal. Teachers will access approved content, apply filters and launch classroom display mode. Admin Staff will manage content records, while authorised reviewers approve content before it becomes visible to Teachers.

4. Non-Functional Requirement Categories

Category	Meaning for this project
Performance	How quickly pages, filters and content launch actions should respond.
Usability	How easy the platform should be for Teachers and Admin Staff to use.
Security	How role access, login and content visibility should be protected.
Compatibility	Which browsers and classroom display environments should be supported.
Availability	When the platform should be available for school usage.
Reliability	How consistently content access and display should work during teaching.
Accessibility	Basic readability and classroom display expectations.
Maintainability	How easily content records, roles and future topics can be maintained.

5. Non-Functional Requirements

NFR ID	Category	Non-Functional Requirement	Priority	Acceptance / Validation Check
NFR-01	Performance	Login page should load within 3 seconds under normal school network conditions.	Must Have	QA checks page load during testing.
NFR-02	Performance	Teacher content library should display results within 3 seconds for normal classroom usage.	Must Have	QA validates library loading with sample content records.
NFR-03	Performance	Class, subject and topic filters should return matching content within 3 seconds for approved content records.	Must Have	QA validates filter response time with sample records.
NFR-04	Performance	Classroom display mode should open within 5 seconds after Teacher clicks Launch Content, excluding large video buffering time.	Should Have	QA validates launch behaviour with sample content.
NFR-05	Usability	Teacher should be able to open approved content from the library in no more than 5 major clicks after login.	Should Have	BA and Teacher validate during UAT.

NFR-06	Usability	Classroom display mode should show a clean view with no unnecessary Admin controls.	Must Have	BA and QA compare display mode against wireframe.
NFR-07	Usability	System messages should be clear when content cannot be opened or no matching content is found.	Should Have	QA validates message scenarios.
NFR-08	Security	Users should access only the screens and actions allowed for their role.	Must Have	QA validates Admin, Teacher and Approval Owner access.
NFR-09	Security	Draft, Under Review and Inactive content should not be visible to Teachers.	Must Have	QA validates visibility rules by content status.
NFR-10	Security	User session should expire after a defined period of inactivity, recommended 30 minutes.	Should Have	Developer confirms session setting and QA validates.
NFR-11	Compatibility	Platform should support latest stable versions of Chrome and Edge browsers.	Must Have	QA performs browser compatibility testing.
NFR-12	Compatibility	Classroom display mode should support common screen resolutions such as 1366 x 768 and 1920 x 1080.	Should Have	QA validates on representative display sizes.

NFR-13	Availability	Platform should be available during school teaching hours, normally 8:00 AM to 4:00 PM on school days.	Must Have	Product Owner confirms operating expectation.
NFR-14	Reliability	System should show a loading message if content takes time to open.	Should Have	QA validates slow-load scenario if testable.
NFR-15	Data Integrity	Mandatory content fields should prevent incomplete records from being submitted for review.	Must Have	QA validates mandatory field rules.
NFR-16	Maintainability	Admin Staff should be able to manage content metadata without Developer support for normal updates.	Should Have	Admin UAT validates content update flow.
NFR-17	Accessibility	Text, buttons and labels should be readable on classroom projector or smart board display.	Should Have	Teacher validates readability during UAT.
NFR-18	Auditability	Approval status should clearly show whether content is Draft, Under Review, Approved or Inactive.	Must Have	QA validates status visibility for authorised users.

6. Role-Based Non-Functional Expectations

Role	Expectation
Teacher	Teacher screens should be simple, quick and suitable for classroom use. Teacher should not see Admin or approval controls.
Admin Staff	Admin screens should support structured data entry and prevent incomplete content records from being submitted.
Principal / Approval Owner	Approval screens should clearly show content status and allow decision making without confusion.
QA Tester	QA should be able to validate role access, browser compatibility, performance and display behaviour using test scenarios.

7. Browser and Display Requirements

Area	Requirement
Browser Support	Latest stable versions of Chrome and Edge should be supported for Release 1.
Screen Support	Classroom display mode should work on projector, smart board or classroom screen.
Display Layout	Display mode should reduce unnecessary buttons and keep learning content visible and readable.
Error Handling	If content cannot open, system should show a clear error message to the Teacher.
Loading Behaviour	If content takes time to open, system should show loading indication.

8. Assumptions

- School classrooms will have internet access during teaching hours.
- Teachers will use supported browsers such as Chrome or Edge.
- Sample content records and content files or links will be available for testing.
- Final performance targets may be refined after technical assessment by the Developer Team.
- Release 1 focuses on teacher-led classroom use only.

9. Dependencies

Dependency ID	Dependency	Owner
DEP-01	Confirmation of supported browsers and classroom devices.	Developer Team / School IT
DEP-02	Availability of sample 3D video, animated video and coding simulation content.	3D / Content Team
DEP-03	Confirmation of school teaching hours for availability expectation.	Principal / Product Owner
DEP-04	Final agreement on content file size and storage approach.	Developer Team / Content Team
DEP-05	UAT availability from Teachers and Admin Staff.	Principal / Product Owner

10. Risks and Mitigation

Risk ID	Risk	Impact	Mitigation
R-01	Large content files may load slowly in classroom.	Teacher may face delay during lesson delivery.	Agree file format, size limit and loading behaviour with Developer and Content Team.
R-02	Unsupported browser or old classroom device may affect display mode.	Content may not display properly.	Confirm browser and device support before UAT.
R-03	Role permissions may be configured incorrectly.	Users may see wrong screens or actions.	Create role-based test scenarios for Admin, Teacher and Approval Owner.
R-04	Teachers may find display mode difficult to use.	Low adoption risk.	Review display mode wireframes and validate with Teachers during UAT.

11. NFR Acceptance Summary

The Non-Functional Requirements will be considered ready when the Product Owner, Developer Team and QA Tester confirm that performance, usability, security, compatibility, availability and display expectations are clear enough for backlog preparation and test planning.

12. Next Step

After NFRD review, the Business Analyst will prepare the Business Rules Document, Use Case Document, Data Field List, Role and Permission Matrix, user stories and acceptance criteria for Sprint 1 and Sprint 2 backlog preparation.